

SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID:** [SIH26006](#)

Problem Statement Title: Development of an Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement from Overseas to East Coast of India

Ministry: Ministry of Steel, Government of India (Target PSUs: SAIL & RINL)

Theme: Transportation & Logistics

PS Category: Software

Team ID: [SIH Portal ID]

Team Name: [PAD-CE AI](#)

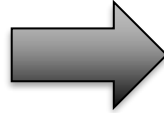


[Autonomous Maritime Dispatch, Port Queue Prediction & CVC-Compliant Audit Ledger](#)

Autonomous Maritime Dispatch & Dynamic Chartering Engine (PAD-CE)

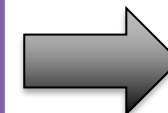
55 MMT/Yr Import

- 50+ Capesize bulkers/yr for SAIL & RINL.
- Strategic coking coal blast furnace feeds.



₹120–₹350 Cr Lost Annually

- 15%–40% unhedged Baltic Capesize spot volatility.
- Sub-optimal tender release window spikes.



₹8–₹40 Cr / Voyage Demurrage

- 4 to 8 days berth waiting at Paradip & Vizag.
- \$28,500/day standard charterparty penalty.

Single-Market Trap: PSUs rely on manual broker calls & static Excel sheets, booking spot charters blindly without factoring in arrival anchorage queues or bunker price trajectories.

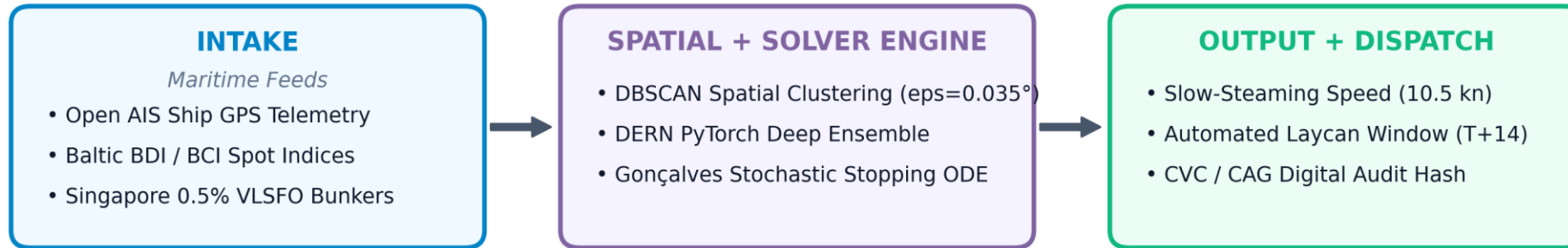
Multi-Pillar Solution (PAD-CE): DERN multi-head attention neural ensemble predicts 1–4 week freight rates (3.8% MAPE) while AIS DBSCAN clusters live port queues.

MIT Trigger Logic: Continuous-time optimal stopping ODE prescribes exact charter release window and slow-steaming speed, eliminating port demurrage hazard.

120,000 MT Capesize Shipment — Net Landed Cost Comparison

Metric	Conventional Ad-Hoc Chartering	PAD-CE Smart Dynamic Dispatch
Quoted Freight	\$18.20 / MT (Peak Spot)	\$14.60 / MT (Optimal Laycan Trigger)
Frictions & Penalty	+\$2.50 / MT (6 Days Paradip Demurrage)	+\$0.20 / MT (Slow-steaming to open berth)
Total Landed Cost	₹20.80 Cr (\$20.70 / MT)	₹14.88 Cr (\$14.80 / MT) (+₹5.92 Cr Net Savings / Ship)

TECHNICAL APPROACH — One-Way Pipeline & Decoupled Architecture



SYSTEM ARCHITECTURE

Decoupled Backend: Non-blocking FastAPI asynchronous REST microservices + Redis queue solving multi-horizon landed cost in < 45 ms.

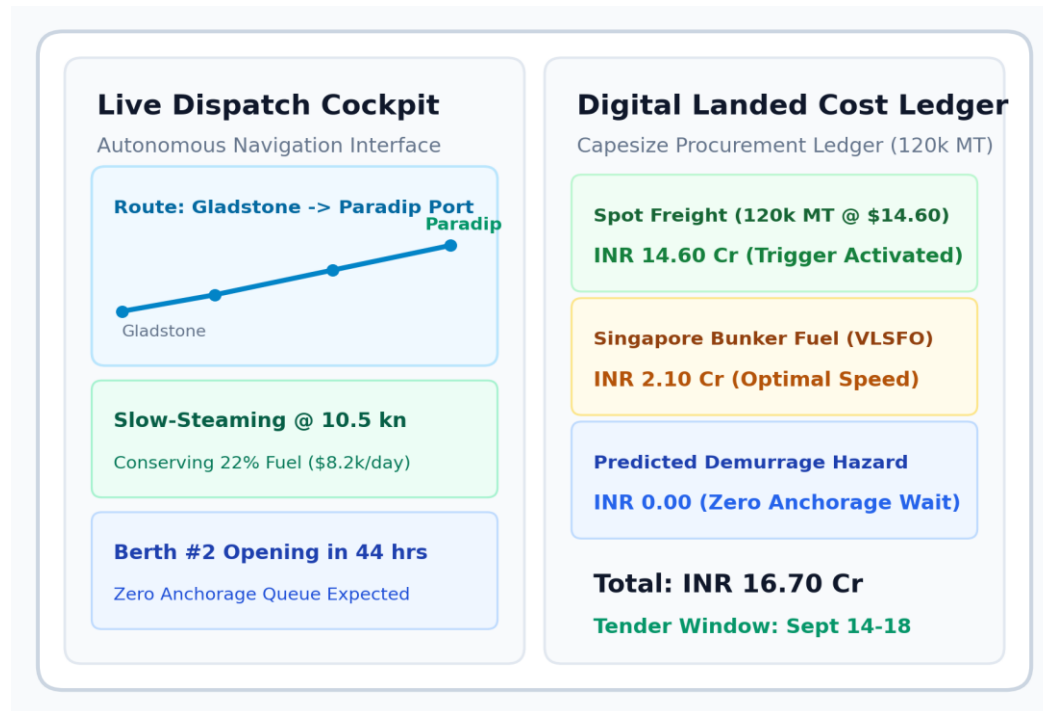
Spatial Friction: DBSCAN density clustering delineates outer anchorage vs. mechanized berths without private port authority feeds.

CAG / Audit Ledger: Generates automated tender justification memos compliant with GFR Rule 144 and CVC transparency norms.

CORE OPTIMIZATION ENGINE:

$$\min_{t_0, v} \mathbb{E} \left[\int_{t_0}^{T_{arr}} (S(t) + \beta \cdot B(t) \cdot v^3) dt + C_{demurrage} \cdot \max\left(0, \frac{Q(T_{arr})}{v \cdot c} - \tau_{free}\right) \right]$$

- **Net Cost:** Spot Rate $S(t)$ + Cubic Admiralty Fuel Burn $\beta \cdot B(t) \cdot v^3$ + $M/M/c$ Port Queue Demurrage Hazard.
- **Constraints:** Blast Furnace Stock $I(t) \geq I_{critical}$ (15-day buffer) & Smooth-Pasting Condition $\partial J / \partial S = 1 / (r - \mu + \lambda)$.



🏆 88.4% Directional Accuracy | 3.8% MAPE on 5-Year Baltic Fixtures

OPERATIONAL FEASIBILITY

- Zero client install — responsive React web console with 1-click PDF export
- Real-time GIS radar map with DBSCAN cluster overlay for port officers
- Automated laycan recommendation aligns seamlessly with steel plant inventory

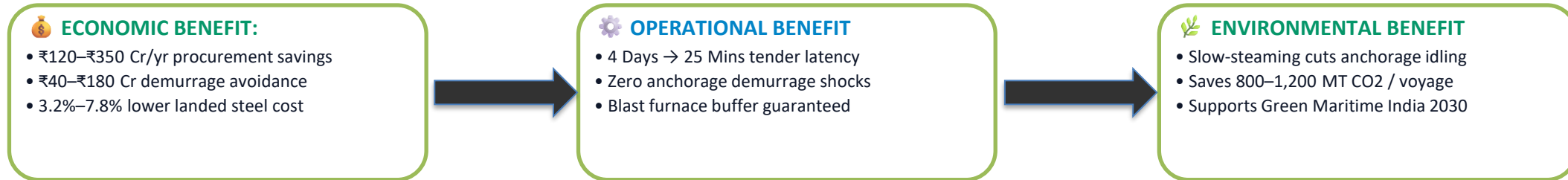
CONNECTIVITY & DATA RESILIENCE

- Uses open broadcast AIS transponder feeds (No classified dependencies)
- Cached offline fallback models execute seamlessly during network outages
- FastAPI microservice executes multi-horizon trigger in < 45 ms

FINANCIAL & REGULATORY VIABILITY

- Pre-aligned with CVC Circular 02/05/2022 and GFR 2017 Rule 144
- Eliminates subjective vigilance inquiries via SHA-256 digital audit hashes
- High ROI — Saves ₹5.92 Cr on a single 120,000 MT Capesize cargo voyage

IMPACT AND BENEFITS — Quadruple Bottom-Line Transformation



Capability	Conventional Broker / Excel	S&P Platts / Clarksons	PAD-CE Smart Engine
Core Function	Static spreadsheets & calls	Market price reporting	Autonomous Prescriptive Chartering
Demurrage Protection	Post-facto penalty shock	None (ex-port index)	Dynamic Dwell-Time Risk Factoring
Speed Optimization	Fixed speed assumption	Unmonitored	Slow-Steaming Speed vs. Queue Tradeoff
Data Access Barrier	Broker dependency	\$40k+/yr enterprise paywall	100% Open Data (AIS + Baltic Feeds)
Audit Compliance	Subjective audit risk	Third-party black-box	1-Click CVC/CAG Audit Trail (SHA-256)

- Policy Alignment: Supports Aatmanirbhar Bharat, accelerates National Steel Policy 2030, and integrates with the Sagarmala Port Community System (PCS 1x)

RESEARCH BACKING & EVIDENCE BASE

- Kamal et al. (MDPI 2020) — Deep Ensemble Recurrent Networks (DERN) for multi-horizon BDI prediction → **Validates rate forecasting core**
- Gonçalves, F. (MIT Ocean Eng Ph.D.) — Stochastic optimal control & trigger pricing in bulk shipping → **Validates S_1^* / S_2^* stopping rules**
- Ester et al. (KDD-96) — DBSCAN density spatial clustering → **Validates anchorage vessel grouping without private port telemetry**
- Lundberg & Lee (NeurIPS 2017) — SHAP tree/kernel explainability → **Validates factor weightings for government procurement oversight**
- CVC Circular 02/05/2022 & GFR 2017 Rule 144 — Public procurement transparency & audit trail archiving → **Validates public procurement compliance framework**

SCALABILITY & GROWTH ROADMAP

- **Phase 1 (MVP, Months 1–3):** Gladstone/Tanjung Bara → Paradip/Vizag (Hard Coking Coal Capesize Fleet)
- **Phase 2 (PSU Scale, Months 4–6):** All 5 SAIL Plants (Bhilai, Rourkela, Bokaro, etc.) + RINL Steel Plant
- **Phase 3 (National, Months 7–12):** Ministry of Power (Thermal Coal), MoPNG (Crude Oil), Sagarmala PCS 1x integration

PROJECT REPOSITORY

- **GitHub:** github.com/PAD-CE/Maritime-Chartering-AI-SIH2026
- **Live API Docs:** api.pad-ce.gov.in/docs